IVAN JORGE CABALLOS ESPINOSA

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PROFILE

Graduate Electrical & Electronic Engineer with hands-on experience in production programming, functional test, and first-line PCB fault finding in an EMS environment. Focused on hardware verification and bring-up: integration, measurement-driven debugging, root-cause isolation, and clear documentation. Targeting Graduate roles in hardware integration, verification, and test/debug.

KEY TECHNICAL SKILLS

- Hardware verification & bring-up: integration, fault isolation (RCA), measurement capture, repeatable documentation
- Production test execution: pass/fail criteria, controlled procedures, traceability mindset, consistent station operation
- PCB debug: visual inspection (placement/solder/BOM), rail checks, continuity/shorts, structured troubleshooting
- Programming & tools: Flash Development Toolkit (Renesas), OPTWIN, MPLAB; bench instruments (PSU, DMMs)
- Embedded platforms: ESP32-S3 (Arduino), I^2C/SPI, Wi-Fi (NTP), ESP-NOW; GitHub/GitHub Pages documentation

PROFESSIONAL EXPERIENCE

Test Technician — Dynamic EMS | Dalgety Bay, Scotland | Nov 2025 – Present

- Program PCB assemblies via station PCs; flash production images and validate configuration using FDT / OPTWIN / MPLAB.
- Execute functional test procedures: power boards using bench PSU at specified conditions, verify outcomes, and capture results for traceability.
- Maintain throughput of ~25–30 boards/day while keeping test execution consistent and records accurate.
- Troubleshoot failed units: visual checks and DMM checks to identify missing rails, shorts, and continuity issues.
- Operational continuity: reconfigured an alternative programming station (PC + auxiliary power board) when a legacy Windows XP station failed.

Royal Mail — Postman (previous / part-time)

Time-critical delivery, reliability, and consistent execution to standard procedures under daily throughput.

EDUCATION

- BSc Electrical & Electronic Engineering — Edinburgh Napier University (Awarded Oct 2023)
- HNC Electrical Engineering — Glasgow Kelvin College

SELECTED PROJECTS (Portfolio)

- Clock and Weather Station — ESP32-S3 (ILI9341 + AHT20 + BMP280, NTP + Open-Meteo, battery-powered): integrated TFT UI + sensing + online forecast; implemented UK NTP time sync (GMT/BST) and on-demand Wi-Fi updates; measured PSU current (~40 mA boot, ~15 mA idle with display on).
- Laser Constant-Current Driver Prototype (0–200 mA target): designed and validated a feedback current driver; documented bench measurements and troubleshooting (including measurement burden effects) in portfolio.
- AI Lab Assistant — Raspberry Pi 4: self-hosted engineering log system (Flask + API-first workflow) for capturing experiments/debug steps and creating searchable lab notes (LAN-only).